Challenge 3.3: Creating Intelligent Cost Database

- Team CODERS

IntelliCost – From Raw POs to Procurement Intelligence

HPCL's procurement history contains thousands of Purchase Orders with valuable price data. However, this data cannot be effectively reused for cost estimation.

The Core Problem

The same item appears across POs with different textual descriptions and no unique identifier.

Example:- "CS Pipe 100 mm"
"Carbon Steel Pipe of 100 mm diameter"
"CS pipeline 100mm SCH40"

Same item. Different text. No unique identity.

Impact on Cost Estimation

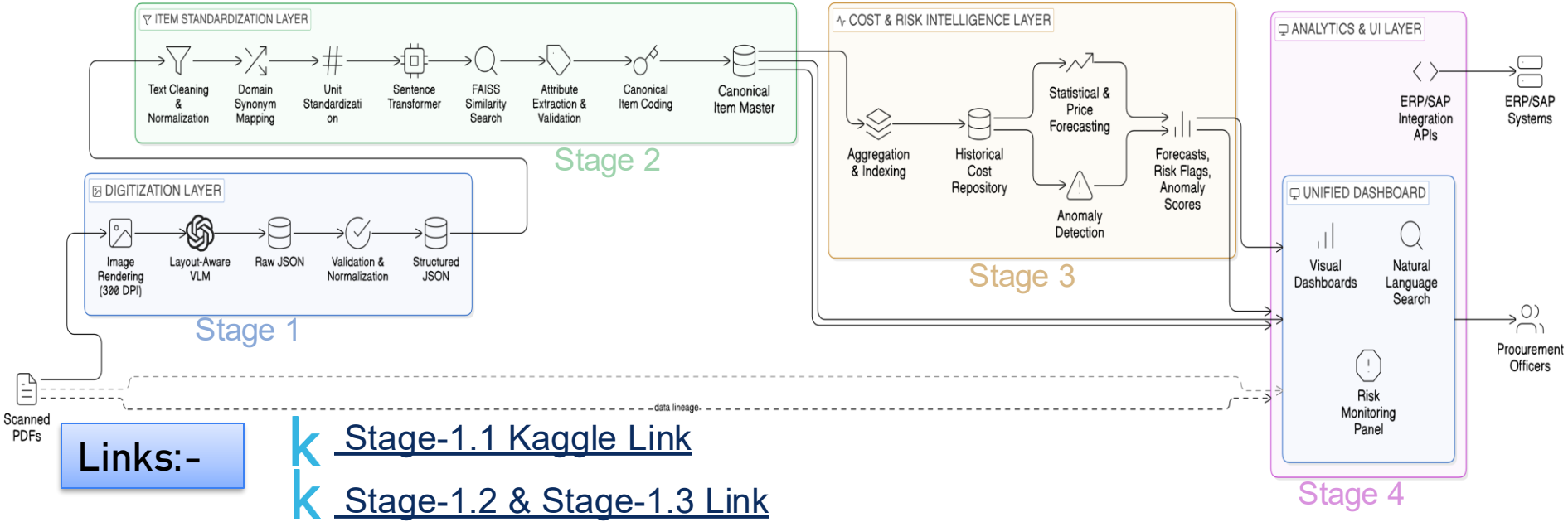
- No reliable item-wise price history
- Manual and inconsistent cost estimation
- Poor vendor and regional comparison
- Broken price trend and variance analysis

Organizations have data, but not decision-ready intelligence.

Our Solution

IntelliCost is built as a modular, end-to-end pipeline that first converts messy purchase orders into reliable structured data before applying analytics. Each stage produces a clear, verifiable output that feeds into the next, ensuring accuracy, scalability, and smooth integration with existing systems. Using AI and NLP techniques, the system identifies identical items from unstructured text, assigns unique item codes, and learns continuously from past and new procurement data to build a standardized cost intelligence database.

Overall IntelliCost Architecture



Links:-

[Stage-1.1 Kaggle Link]

[Stage-1.2 & Stage-1.3 Link]

[Stage 2,3,4 Github Repository Link]

[Streamlit Live Link]

Stage 1 – Digitizing Messy Purchase Orders

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1.1 PDF → Image

First, each page of the PO is converted into an image to ensure that no visual information such as stamps or handwritten notes is lost.

Technologies

- ❑ 300 DPI
- ❑ Poppler PDF Rendering

1.2 Image → Raw JSON

Each PO page image is processed using a layout-aware **Vision-Language model**. The page is split into header and body sections to improve extraction accuracy. All visible text and tables are captured and stored page-wise in raw JSON format.

Technologies

- ❑ Vision-Language Model
- ❑ Layout Aware Extraction
- ❑ Hallucination Control

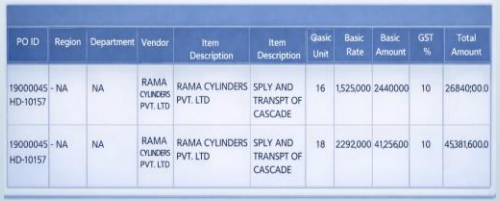
1.3 Raw JSON → Structured JSON + Spreadsheet

In Step-3, these fields are **standardized** into consistent numeric and categorical formats to enable reliable computation, validation and tabular analysis.

Eg:- "7,74,74,000" → 77474000
"18%" → 18

Technologies

- ❑ Schema Validation
- ❑ Type-safe casting
- ❑ Regex normalization



PO ID	Region	Department	Vendor	Item Description	Item Description	Basic Unit	Basic Rate	Basic Amount	GST %	Total Amount
19000045 - HD-10157	NA	NA	RAMA CYLINDERS PVT. LTD	RAMA CYLINDERS PVT. LTD	SPLY AND TRANSPT OF CASCADE	16	1,525,000	2440000	10	26840,000.0
19000045 - HD-10157	NA	NA	RAMA CYLINDERS PVT. LTD	RAMA CYLINDERS PVT. LTD	SPLY AND TRANSPT OF CASCADE	18	2292,000	4125600	10	45381,600.0

Stage 1 converts unstructured, scanned POs into reliable structured data. This eliminates noise and makes the data safe for downstream analytics.

Stage 2 – Intelligent Item Standardization

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2.1 Raw Item Text → Normalized Text

Procurement item descriptions in POs often vary in abbreviations, spelling, naming conventions, and units. In this step, raw item text is cleaned and normalized into a consistent format, removing noise while preserving the original engineering and material intent.

Technologies

- Regex-based text cleaning and normalization
- Domain-specific synonym mapping (CS, MS, SS, GI, etc.)
- Unit, dimension, and material standardization rules

2.2 Normalized Text → Semantic Item Matching

Normalized item descriptions are converted into semantic representations and compared against historical procurement data. This allows the system to group items that are written differently but refer to the same underlying material or component.

Technologies

- Sentence-transformer based text embeddings
- FAISS-based vector similarity search
- Cosine similarity scoring for match confidence

2.3 Semantic Match → Canonical Item Code

For engineering-critical items, key attributes such as diameter, thickness, and size are extracted and validated using strict rules. If a safe match exists, the item is linked to the corresponding canonical HPCL item code; otherwise, a new code is created and added to the item master.

Technologies

- Numeric attribute extraction from text
- Hard-constraint validation logic
- Persistent item master (CSV + FAISS index)

Stage 2 transforms inconsistent item descriptions into a single, reliable canonical item master, ensuring semantic consistency and engineering safety for all downstream analysis.

Stage 3 – Cost & Risk Intelligence Models

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3.1 Canonical Item → Historical Cost Repository

All purchase order prices are mapped to canonical item codes to ensure consistent tracking of the same item across vendors and POs. The data is consolidated across locations, suppliers, and time periods to create a clean and reliable historical cost repository for further analysis.

Technologies

- ❑ Vendor and region-wise grouping
- ❑ Pandas-based price aggregation
- ❑ Time-series indexing

3.2 Historical Prices → Price Forecasting

Historical price data is analyzed to identify trends, seasonality, and normal price ranges for each canonical item. Based on these patterns, future price ranges are forecasted to support budgeting, tender planning, and informed vendor negotiations.

Technologies

- ❑ Statistical time-series forecasting models
- ❑ Trend smoothing and controlled extrapolation
- ❑ Confidence band estimation

3.3 Price Data → Anomaly Detection

Price data is continuously monitored to identify abnormal behavior such as sudden spikes, unusual vendor pricing, or unexpected deviations from historical patterns. These signals help procurement teams detect pricing risks and potential cost leakages early.

Technologies

- ❑ Statistical deviation analysis
- ❑ Z-score based anomaly detection
- ❑ Rule-driven alert thresholds

Stage-3 transforms standardized procurement data into actionable insights, enabling cost optimization, vendor transparency, and proactive risk detection.

Stage 4 – Unified Cost Intelligence Dashboard

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4.1 Item-Centric Analytics Interface

Users search for items using natural language, which is resolved through the Stage-2 standardization layer into canonical item codes. All cost, vendor, and risk analytics are presented consistently at the canonical item level, ensuring that every item has a single, unified analytical view across purchase orders, suppliers, and time periods.

Technologies

- ❑ Streamlit interactive UI
- ❑ Model-to-UI integration
- ❑ Dynamic filters and search

4.2 Visual Cost & Vendor Intelligence

Historical price trends, vendor-wise comparisons, and forecast outputs are visualized in a clear and intuitive manner. These views allow users to quickly understand pricing behavior, compare vendors, and identify cost patterns without requiring manual data analysis or spreadsheet-based effort.

Technologies

- ❑ Time-series visualizations
- ❑ Vendor comparison tables
- ❑ Interactive charts

4.3 Risk & Anomaly Monitoring Panel

Anomalous purchase orders and vendor-level risks identified by analytical models are surfaced in a dedicated monitoring view. Each alert is accompanied by contextual information and remains explainable and traceable, supporting audit, compliance, and informed follow-up actions.

Technologies

- ❑ Anomaly model outputs
- ❑ Tabular risk views
- ❑ Explainable alert logic

Stage 4 brings analytics to the user through a unified dashboard, converting model outputs into clear, explainable, and decision-ready procurement insights.

Assumptions, Challenges & Future Goals

Assumptions & Challenges

1) Synthetic Historical Data
Due to data confidentiality constraints, historical procurement data was synthetically generated. While this allowed validation of end-to-end logic, real-world scale and noise patterns are yet to be tested on actual HPCL data.

2) OCR Strategy Trade-off
Stage-1 uses a Vision-Language OCR pipeline that extracts the entire page image into raw JSON, rather than directly generating structured fields. This ensures completeness but increases post-processing complexity and latency.

3) Pipeline Decoupling
Currently, Stage-1 (Digitization) and Stage-2/3/4 (Intelligence & Dashboard) are executed as separate pipelines. A fully automated, real-time data flow is planned but not yet integrated, including division level metadata when available.

4) Fixed FAISS Confidence Threshold
Semantic item matching currently uses a static similarity threshold. While effective, it may not adapt optimally across all item categories and description complexities.

Future Goals

1) Deployment on Real HPCL Procurement Data
Validate and refine the system using actual historical purchase order data to improve robustness, accuracy, and confidence calibration at real operational scale.

2) End-to-End Pipeline Automation
Integrate Stage-1 digitization directly with downstream intelligence layers to enable a fully automated, near real-time data ingestion and processing pipeline.

3) Adaptive Item Matching Logic
Introduce dynamic similarity thresholds for semantic matching based on item category, description complexity, and historical match confidence to improve precision.

4) Enhanced Attribute, Reason & Context Inference
Extend attribute extraction to infer missing contextual fields such as department, project, and region/division using rule-based and NLP-driven heuristics.

5) Audit & Compliance–Ready Reporting
Generate traceable, exportable reports that capture item-level pricing history, anomalies, and decision rationale to support audit reviews and regulatory compliance.

References

1) HPCL-Annual-Report-2024-25
2) https://www.scribd.com/document/672782594/HPCL-PO
3) https://www.scribd.com/document/725622727/work-order-hpcl
4) https://bidplus.gem.gov.in/bidding/bid/documentdownload/7519404/1739438327.pdf
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